AI Exam Answer Grader

A Reusable NLP Asset for Rubric-Based Descriptive Answer Assessment

Week 4 Deliverable — Baseline vs Improved Model Comparison | Target: 26 Sep – 2 Oct 2026

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1. Summary

Week 3 delivered Version 1 of the improved pipeline: TF-IDF, sentence-embedding and cross-encoder models evaluated with 5-fold cross-validation grouped by question. Week 4 compares the improved models against the baseline in depth, using the saved out-of-fold predictions (every answer is predicted by a model that never saw its question). It adds one new method (an ensemble), a significance test, an analysis by answer quality, weak-answer detection, a check of the confidence score, and a per-question comparison.

Main result: the fine-tuned cross-encoder (B3) reduces the mean absolute error from 2.49 marks (Week 1–2 baseline) to 0.63 marks. Its gain over the calibrated TF-IDF baseline is 0.09 marks, and a bootstrap test over questions confirms this is a real improvement (95% CI 0.03 to 0.15). It also halves the error on weak answers compared with the calibrated baseline, and its confidence score reliably identifies the predictions most likely to be wrong.

2. Evaluation Setup

- **Data:** Mohler short-answer grading dataset, 2,264 answers to 79 computer science questions after removing 9 corrupted answers; human mark = average of two graders (0–5). Mean mark 4.17.
- **Protocol:** 5-fold cross-validation grouped by question (15–16 questions per fold), so all results are on unseen questions. Results are mean ± standard deviation across folds.
- **Metrics:** MAE, RMSE, agreement within ±1 mark and Pearson correlation (handout requirements), plus a bootstrap significance test, precision/recall for weak-answer detection, and error by confidence group.
- **Consistency check:** the Week 4 notebook recomputes every Week 3 metric from the saved predictions; the largest difference from the Week 3 results file is 0.

3. Methods Compared

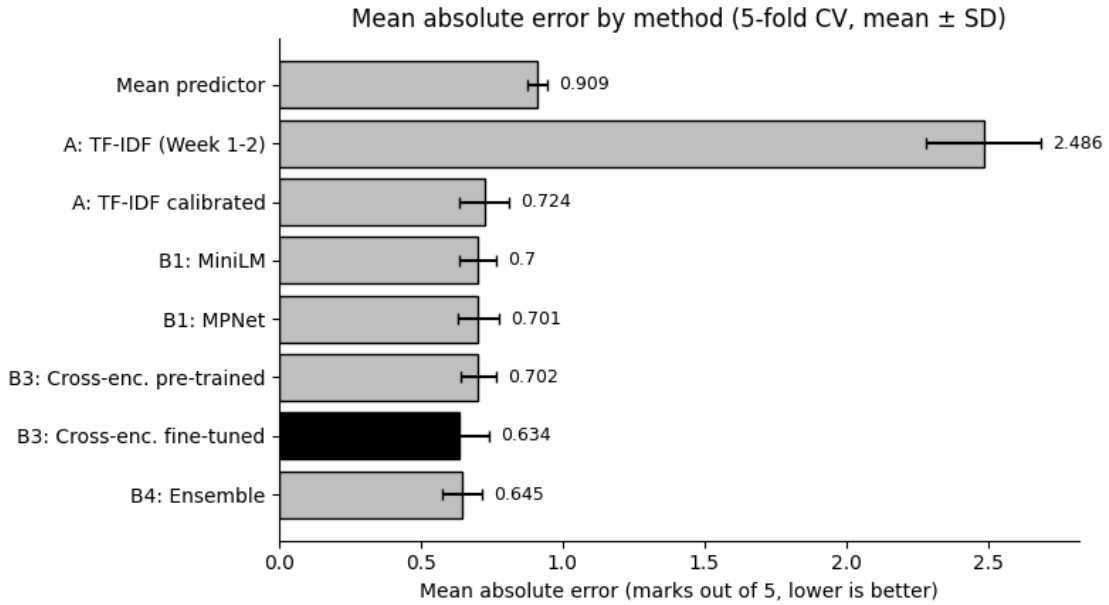
Label	Method
Floor	Mean predictor: always predicts the average training mark.
A (Week 1–2)	Baseline: TF-IDF cosine similarity + keyword concept coverage, fixed linear scaling to marks.
A (calibrated)	Same TF-IDF features, mapped to marks by a regression learned on the training questions.
B1 MiniLM / MPNet	Sentence-embedding similarity (all-MiniLM-L6-v2 / all-mpnet-base-v2) + concept coverage, calibrated.
B3 pre-trained	Cross-encoder (stsb-TinyBERT-L4) similarity + concept coverage, calibrated.
B3 fine-tuned	The same cross-encoder fine-tuned in each fold to predict the human mark directly.
B4 ensemble (new)	Equal-weight average of B3 fine-tuned and B1 MiniLM. Weights fixed in advance, not tuned on test data.

Label B2 is reserved for the LLM rubric-grounded grader, planned for Week 5.

4. Main Comparison

Method	MAE ↓	RMSE ↓	Within ±1 (%) ↑	Pearson r ↑
Floor: Mean predictor	0.909 ± 0.036	1.115 ± 0.093	79.9 ± 5.5	–
A: TF-IDF (Week 1–2)	2.486 ± 0.204	2.795 ± 0.182	15.0 ± 5.6	0.467 ± 0.067
A: TF-IDF (calibrated)	0.724 ± 0.087	0.975 ± 0.095	72.4 ± 5.0	0.486 ± 0.071
B1: MiniLM (calibrated)	0.700 ± 0.066	0.936 ± 0.075	74.3 ± 2.2	0.543 ± 0.043
B1: MPNet (calibrated)	0.701 ± 0.072	0.938 ± 0.081	74.4 ± 3.2	0.537 ± 0.053
B3: Cross-encoder pre-trained	0.702 ± 0.064	0.944 ± 0.076	75.7 ± 4.0	0.531 ± 0.063
B3: Cross-encoder fine-tuned	**0.634 ± 0.108**	**0.923 ± 0.127**	**75.5 ± 5.7**	**0.589 ± 0.099**
B4: Ensemble (B3 fine-tuned + B1 MiniLM)	**0.645 ± 0.069**	**0.876 ± 0.090**	**77.3 ± 4.9**	**0.615 ± 0.071**

Pearson correlation is undefined for the mean predictor because its predictions are constant. Folds in which each method beats calibrated TF-IDF on MAE: B1 MiniLM 4/5, B1 MPNet 5/5, B3 pre-trained 4/5, B3 fine-tuned 4/5, B4 ensemble 5/5.



B3 fine-tuned has the lowest MAE. The new B4 ensemble has the best RMSE, ±1 agreement and correlation, and is noticeably more stable across folds (MAE ± 0.069 instead of ± 0.108), because averaging with the stable MiniLM model smooths out the cross-encoder's worst mistakes.

5. Is the Improvement Real? — Bootstrap Test

With 79 questions, a small MAE difference can occur by chance. Questions (not individual answers, since answers to one question are strongly related) were resampled with replacement 2,000 times, and both methods' MAE recomputed each time. A positive gain means the first method is better; if the 95% confidence interval excludes 0, the difference is unlikely to be chance.

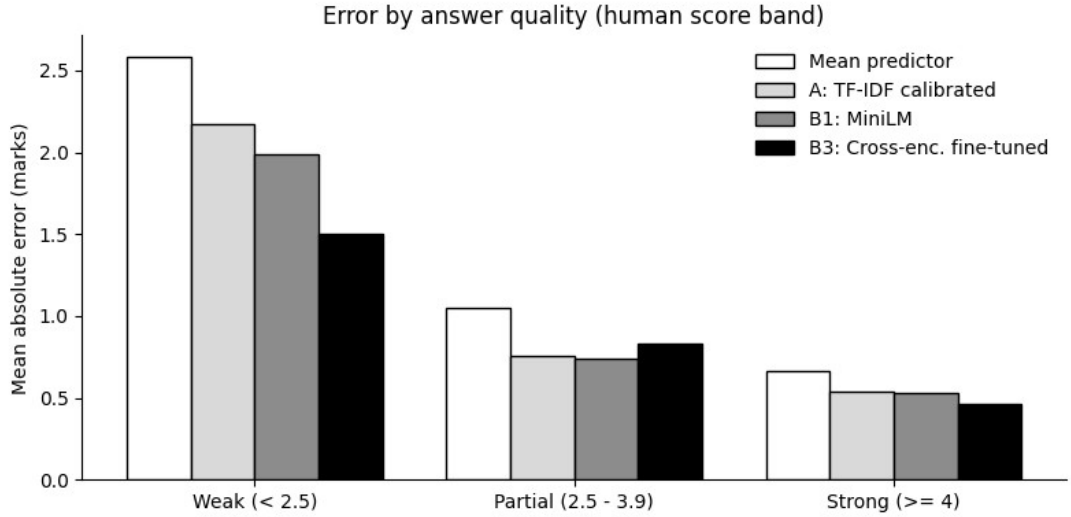
Comparison	MAE gain	95% CI	First better (%)	Verdict
B3 fine-tuned vs A: TF-IDF (Week 1–2)	1.855	[1.686, 2.033]	100.0	Real improvement
B3 fine-tuned vs Mean predictor	0.276	[0.204, 0.343]	100.0	Real improvement
B3 fine-tuned vs A: TF-IDF calibrated	0.091	[0.034, 0.146]	99.9	Real improvement
B3 fine-tuned vs B1 MiniLM	0.068	[0.014, 0.120]	99.1	Real improvement
B1 MiniLM vs A: TF-IDF calibrated	0.024	[0.003, 0.044]	99.1	Real improvement
B4 ensemble vs B3 fine-tuned	−0.012	[−0.039, 0.017]	20.8	Not significant
B4 ensemble vs A: TF-IDF calibrated	0.079	[0.046, 0.111]	100.0	Real improvement

Every improved method is significantly better than the baseline. Fine-tuning adds a real gain on top of pre-trained embeddings. The embedding gain over calibrated TF-IDF is statistically real but very small (0.024 marks). The ensemble and B3 fine-tuned are not significantly different on MAE.

6. Results by Answer Quality

Most answers score 4–5, so overall averages are dominated by good answers. The errors were therefore split by the human mark: weak (below 2.5, 195 answers), partial (2.5–3.9, 465 answers) and strong (4 and above, 1,604 answers).

Method	Weak (< 2.5)	Partial (2.5–3.9)	Strong (≥ 4)
Floor: Mean predictor	2.585	1.052	0.661
A: TF-IDF (Week 1–2)	1.100	2.229	2.729
A: TF-IDF (calibrated)	2.172	0.760	0.535
B1: MiniLM (calibrated)	1.990	0.740	0.529
B1: MPNet (calibrated)	2.002	0.745	0.529
B3: Cross-encoder pre-trained	2.030	0.756	0.523
B3: Cross-encoder fine-tuned	**1.499**	**0.832**	**0.467**
B4: Ensemble	1.736	0.743	0.481



This is where the improved model matters most. On weak answers the calibrated baseline is off by 2.17 marks and the mean predictor by 2.59, because both push every answer towards the high average. B3 fine-tuned reduces this to 1.50. It is also best on strong answers, but slightly worse on partial answers (0.83),

where the ensemble is better (0.74). The Week 1–2 baseline only looks good on weak answers because it scores almost every answer low.

7. Can the Grader Catch Weak Answers?

A practical use for faculty is to flag answers that probably deserve less than 3 out of 5 for a careful manual check. An answer is flagged when the predicted score is below 3. 313 of the 2,264 answers really scored below 3.

Method	Flagged	Precision	Recall	F1
Floor: Mean predictor	0	0.000	0.000	0.000
A: TF-IDF (Week 1–2)	1,871	0.166	0.994	0.285
A: TF-IDF (calibrated)	0	0.000	0.000	0.000
B1: MiniLM (calibrated)	35	0.686	0.077	0.138
B1: MPNet (calibrated)	41	0.683	0.089	0.158
B3: Cross-encoder pre-trained	30	0.733	0.070	0.128
B3: Cross-encoder fine-tuned	**237**	**0.430**	**0.326**	**0.371**
B4: Ensemble	140	0.507	0.227	0.313

The calibrated models almost never predict below 3: a regression trained on mostly high marks pulls every prediction towards the average, so they miss over 90% of weak answers, and calibrated TF-IDF flags none at all. The fine-tuned cross-encoder is the only model that catches a useful share (one in three weak answers, F1 0.37). This is still the main weakness of the system and a key target for Week 5.

8. Does the Confidence Score Work?

The Week 3 pipeline reports confidence = 1 − |cross-encoder score − MPNet score| ÷ 5. The answers were split into four equal groups by confidence:

Confidence group	Confidence range	Answers	MAE	Errors > 1.5 marks
Lowest 25%	0.40 – 0.86	566	0.887	21.0%
Lower-middle 25%	0.86 – 0.93	566	0.647	11.0%
Upper-middle 25%	0.93 – 0.98	566	0.607	8.5%
Highest 25%	0.98 – 1.00	566	0.384	5.5%

Error falls steadily as confidence rises (correlation −0.26). Answers in the lowest-confidence quarter are almost four times as likely to be off by more than 1.5 marks as those in the highest. Sending the lowest-confidence 25% of answers to faculty review would therefore catch a large share of the serious errors, which supports the human-in-the-loop design.

9. Per-Question Comparison

B3 fine-tuned beats calibrated TF-IDF on 50 of the 79 questions. The five questions where it is least accurate are those whose correct answers need domain knowledge or allow many valid phrasings:

- How many steps does it take to search a node in a binary search tree? (MAE 1.62; students correctly answer "log n" while the reference says "the height of the tree")
- How are objects initialized when they are created? (MAE 1.47)
- What is the base case for a recursive implementation? (MAE 1.42)
- How are overloaded functions differentiated by the compiler? (MAE 1.38)
- What is the difference between a static and a non-static array? (MAE 1.33)

On all five, calibrated TF-IDF is more accurate, which means the fine-tuned model is less reliable on questions whose accepted answers differ most from the reference.

10. Conclusions

1. **The improved approach clearly beats the baseline.** MAE falls from 2.49 to 0.63 marks, correlation rises from 0.47 to 0.59, and the gain over the calibrated baseline is statistically significant.
2. **Fine-tuning is what makes the difference.** Pre-trained embeddings give only a very small gain over calibrated TF-IDF; fine-tuning the cross-encoder on human marks gives a significant further gain.
3. **B3 fine-tuned remains the pipeline's scoring model.** It has the lowest MAE and is the only model that detects a useful share of weak answers. The B4 ensemble is a strong alternative (more stable, best correlation) and will be re-evaluated in Week 6.
4. **Main remaining weakness: weak answers.** All models under-detect weak answers because the training data is dominated by high marks, and they struggle with correct answers that need domain knowledge. Both require judging correctness rather than similarity, which motivates the LLM rubric-grounded grader (B2) in Week 5.
5. **The confidence score is meaningful.** It can be used to route uncertain answers to faculty review.

11. Next Steps (Week 5 — Evaluation & Error Analysis)

- **Approach B2:** evaluate the LLM rubric-grounded grader on the same folds, focusing on weak and partial answers.
- **Class dataset:** build a labelled test set from our own anonymized mid-term answers, with rubrics prepared with faculty.
- **Error analysis:** categorise 10 or more representative errors across methods, as required by the handout.
- **Weak-answer handling:** test re-weighting low-scoring answers during fine-tuning so the model predicts low marks more often.

Code and results: Group 2 AI Exam Grader Week 4.ipynb; results folder (week 4 comparison table, bootstrap tests, mae by answer quality, weak answer detection, confidence check, per question comparison, and three charts).